Type MS · Describe transcription and translation in human cells

Answer 1 of 5 · Question 8 part a.ii · 5 marks

2023 JUN · Paper 02 (regional R) · find part a.ii in this mark scheme extract

8	(a) (i)	Any two from • DNA/RNA/genetic material; • surrounded by a protein coat/capsid; • (protein) spikes on outer surface;		2
	(ii)	Any five from • transcription (of DNA); • into mRNA; • translation (of mRNA) at the ribosome; • by tRNA; • reference to (tRNA) anticodon pairing with (mRNA) codon; • amino acids assembled to form (viral) proteins;		5
	(b) (i)	Human Immunodeficiency virus/HIV;		1
	(ii)	• (1400000 – 650000) = 750000; • (750000 ÷ 1400000) (x 100) = 54%	Allow any figure that rounds to 54	2

Total for Question 8 = 10 Marks

Type MS · Describe transcription and translation in human cells

Answer 2 of 5 · Question 6 part d · 6 marks

2019 JUN · Paper 01 (regional R) · extract 1 of 2 · find part d in this mark scheme extract

6(a)	• holds DNA/chromosomes/genetic information in nucleus; • controls what enters and leaves nucleus / allows movement of substances/named substance out of/into the nucleus/exchange of substances between nucleus and cytoplasm;		**(2)**

Question Number	Answer	Additional guidance	Mark
6(b)	B CCG;		**(1)**

Question Number	Answer	Additional guidance	Mark
6(c)	threonine;		**(1)**

Question Number	Answer	Additional guidance	Mark
6(d)	Any six from: • tRNA has anticodons; • translation; • of mRNA; • amino acids carried by tRNA; • joined together; • by peptide bonds; • forming polypeptide/protein; • (final) protein released;	Mp3 must be in context of translation	**(6)**

Question Number	Answer	Additional guidance	Mark
6(e)	• causes change in base sequence/genetic code/named type of mutation; • of DNA;		**(2)**

Total 12 Marks

Type MS · Describe transcription and translation in human cells

Answer 2 of 5 · Question 6 part d · 6 marks

2019 JUN · Paper 01 (regional R) · extract 2 of 2 · find part d in this mark scheme extract

Paper 1: Subject Human Biology

Question Number	Answer	Additional guidance	Mark

Type MS · Describe transcription and translation in human cells

Answer 3 of 5 · Question 9 part a.iii · 3 marks

2020 JAN · Paper 02 · find part a.iii in this mark scheme extract

Number				
9	(a) (i)	Any three from: • amino group removed/broken down(1) • in liver(1) • by deamination(1) • to produce urea (1)	Allow description of amino group e.g. nitrogenous group	3
	(ii)	Any three from: • less protein broken down by protease(1) • to amino acids 1) • in stomach/small intestine(1) • reference to reduced amount of phenylalanine/amino acids (in low protein diet)(1)		3
	(iii)	• reduced number of amino acids available(1) • during translation(1) • on ribosomes(1) • incorrect/less protein formed(1)	Allow less translation Allow less protein synthesis	3
	(b) (i)	heterozygous/Nn(1)		1
	(ii)	100% (1)		1
	(iii)	• both parents carry recessive allele/allele for PKU/heterozygous(1) • person 3 has inherited one recessive allele from each parent(1) • person 3 has PKU/homozygous recessive(1)	Reject any reference to sex-linked	3

Total for Question 9 = 14 marks

Type MS · Describe transcription and translation in human cells

Answer 4 of 5 · Question 5 part c.i · 3 marks

2022 JUN · Paper 02 (regional R) · extract 1 of 2 · find part c.i in this mark scheme extract

5 (a)	• two monosaccharides; • joined together;		2
(b)	molecule too large;		1
(c) (i)	• transcription (of lactase gene) does not occur; • RNA is not produced/synthesised; • lactase/protein/enzyme not produced;		3
(ii)	• lactose can't be digested; • into components to be absorbed;		2
(d)	• they eat more dairy produce; • being able to digest it would give them an advantage;		2
(e)	• water passes from cells; • into gut; • diarrhoea causes poor absorption of vitamin D and Ca^{2+}; • as they are water soluble;		4

Total 14 marks

Type MS · Describe transcription and translation in human cells

Answer 4 of 5 · Question 5 part c.i · 3 marks

2022 JUN · Paper 02 (regional R) · extract 2 of 2 · find part c.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Describe transcription and translation in human cells

Answer 5 of 5 · Question 2 part a.iii · 1 mark

2023 JUN · Paper 02 (regional R) · find part a.iii in this mark scheme extract

2 (a)	(i)	in the following order: • proteases; • protein Q; • amino acids;	Reject protein P	3
	(ii)	stomach/ small intestine/ pancreas/ duodenum/ ileum;		1
	(iii)	DNA / mRNA / messenger RNA;		1
(b)	(i)	• as pH increases activity increases; • to pH2/peaks at pH2/to optimum; • decreases from pH2;	Allow increases and then decreases for one mark	3 exp
	(ii)	Any two from: • enzyme is denatured (at pH2); • change in shape of active sites; • no enzyme-substrate complexes formed/no binding with substrate;		2 exp
	(iii)	• repeat; • calculate a mean/average/remove anomalous result		2 grad
	(iv)	Any one from: • temperature; • substrate concentration/volume; • enzyme concentration/volume;		1 grad

Total for Question 2 = 13 Marks